

From Revert to Remedy: Process Modeling Wikipedia’s Dispute Escalation

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Abstract—Wikipedia operates as a self-governing encyclopedia in which volunteer editors resolve conflict through a five-stage escalation pathway, from talk page discussion to binding arbitration. We formalize this pathway as a Business Process Model and Notation (BPMN 2.0) swimlane model, mapping participants, gateways, and remedial outcomes across all five intervention venues. The model is grounded in a structured dataset of approximately 170 Arbitration Committee (ArbCom) cases spanning 2004–2023, collected via a purpose-built pipeline against the MediaWiki, ORES, Pageviews, and XTools APIs. An interactive dashboard renders each case as a navigable BPMN diagram alongside analytics covering case duration, statement distributions, party types, and remedy verbs. Across five high-profile sample cases, we find that bilateral edit-war pairs drive the majority of reverts, that over 60% of ArbCom participants are traceable to earlier ANI threads, and that abuse-filter hit counts reliably distinguish sanctioned from unsanctioned editors. The process model and dataset are released publicly to support reproducible research on collaborative online governance.

Index Terms—Wikipedia, BPMN, process modeling, dispute resolution, arbitration, edit wars, collaborative governance, escalation prediction, MediaWiki API

I. INTRODUCTION

Every hour, editors on the English Wikipedia open thousands of disputes. Most die quietly on article talk pages. A few harden into edit wars, climb through noticeboards, and end in binding sanctions from the Arbitration Committee (ArbCom), Wikipedia’s court of last resort. Despite decades of study, no prior work has given this escalation a formal process model, leaving practitioners and researchers without a shared, executable representation of how conflict moves through the system.

We address this gap with three contributions: (1) a BPMN 2.0 swimlane model of the complete dispute escalation pathway; (2) a structured dataset of ~170 ArbCom cases that grounds the model in 20 years of real cases; and (3) an interactive dashboard [17] that renders each case as a navigable BPMN diagram alongside analytics covering duration, parties, and remedies. The process model makes implicit governance logic explicit: gateways represent policy decisions, swimlane

boundaries mark jurisdictional transitions, and timer events capture escalation deadlines.

II. BACKGROUND

A. Wikipedia’s Governance Structure

Wikipedia’s English edition is self-governed by a layered hierarchy: *editors* (any contributor), *administrators* (elected, can block and protect), *bureaucrats* (grant admin rights), and the *Arbitration Committee* (ArbCom), an annually elected body that issues binding conduct sanctions [1]. Three core content policies (Neutral Point of View, Verifiability, and No Original Research) define the terrain of most disputes.

B. The Five-Stage Escalation Pathway

Wikipedia employs *graduated intervention*: binding force increases as lower-level mechanisms fail [3]. Fig. 1 shows our BPMN swimlane model of this pathway, with lanes separating the four participant tiers and XOR gateways encoding the “resolved vs. escalate” policy decision at each stage.

Stage 1: Talk Page. Editors must attempt resolution directly on the article talk page before escalating. The DRN requires evidence of discussion spanning at least two days [4].

Stage 2: Third Opinion / RfC. A Third Opinion involves exactly two disputants and a volunteer referee. Requests for Comment (RfC) bring in broader community input for multi-party or policy disputes.

Stage 3: Dispute Resolution Noticeboard (DRN). Volunteer moderators facilitate *content* disputes only. Cases close as consensus, referral, or unresolved [4].

Stage 4: Administrators’ Noticeboard Incidents (ANI). *Conduct* violations (harassment, sock-puppetry, edit warring) enter the admin tier here. Administrators may sanction without formal proceedings.

Stage 5: Arbitration Committee (ArbCom). The court of last resort for conduct. Cases span five structured subpages (main, evidence, workshop, proposed decision, remedies). Decisions are binding: site bans, topic restrictions, supervised editing.

C. BPMN as a Formalism for Online Governance

Business Process Model and Notation 2.0 (BPMN) is an ISO/IEC standard graphical language for specifying workflows [18]. Its constructs map naturally onto Wikipedia’s

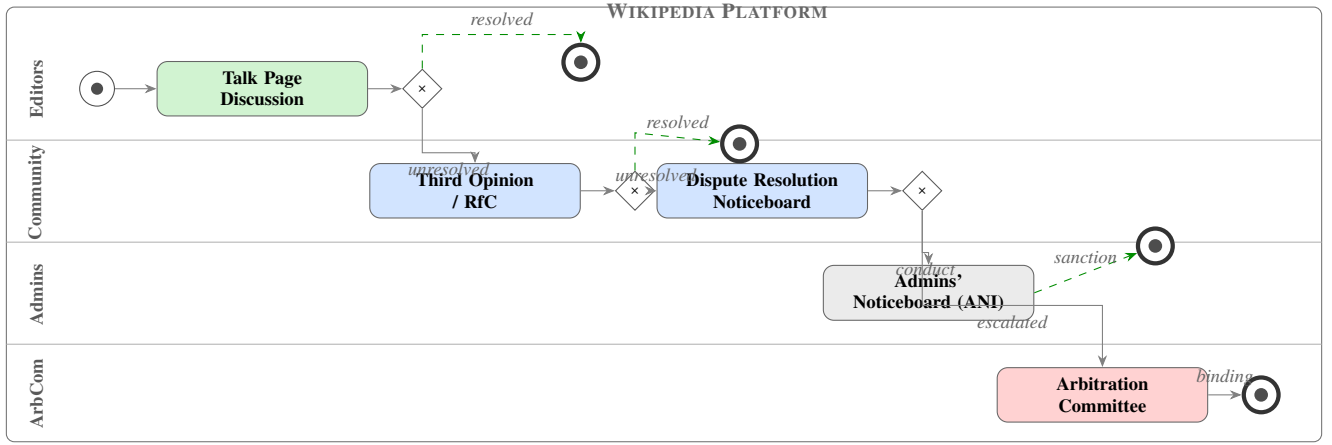


Fig. 1. BPMN 2.0 swimlane model of Wikipedia’s dispute escalation. Lanes separate participant tiers; $\times$ -gateways encode the “resolved vs. escalate” policy decision at each stage. Dashed green flows indicate resolved exits; solid flows indicate escalation. Start and end events follow BPMN 2.0 notation.

governance: *tasks* correspond to discussion and decision activities; *swimlanes* encode jurisdictional boundaries (editors, community volunteers, administrators, committee); *XOR gateways* represent binary policy decisions (“resolved?”); and *end events* distinguish voluntary resolution from imposed remedy. Formalizing the process this way exposes structures (gateway fan-out, inter-lane message flows, timer boundary events for escalation deadlines) that informal narrative descriptions obscure.

III. RELATED WORK

A. Process Mining and Online Governance

Process mining extracts process models from event logs [20]. Applied to Wikipedia, revision histories constitute precisely such logs: each revision is a timestamped event with actor, artifact, and action. Prior work has not applied formal process mining tools to the dispute resolution pathway; we take a step in this direction by constructing BPMN models from case subpages and validating them against collected event logs.

B. Edit War Detection

Kittur et al. [5] introduced foundational metrics for detecting conflict through revert patterns, showing that mutual reversion sequences reliably identify edit wars. In subsequent work, Kittur and Kraut [6] extended this analysis to coordination breakdowns in online production groups more broadly. Yasseri et al. [7] quantified edit wars globally using a *mutually reverting edit index* (M-index), finding that politically, religiously, and geographically contentious topics consistently attract the most conflict.

C. Dispute Resolution Mechanisms

Wikipedia’s own documentation describes a graduated series of venues through which disputes are expected to travel [2]. Geiger and Ribes [9] examined Wikipedia’s anti-vandalism infrastructure through an ethnographic lens, documenting how bots and human editors collaborate in governance. Taraborelli

and Ciampaglia [10] analyzed Articles for Deletion (AfD) as a governance mechanism, providing a model for studying how consensus forms under conflict. Halfaker et al. [8] documented the hardening of Wikipedia’s newcomer environment, connecting dispute resolution stringency to editor retention decline. Relatedly, Halfaker et al. [15] showed that reverts disproportionately discourage new contributors, a dynamic visible in the dispute cases we study.

D. Network Analysis of Conflict

Kumar et al. [11] proposed methods for detecting conflict between user pairs through interaction network analysis. Welser et al. [12] identified distinct editor roles through behavioral signatures, a framework applicable to characterizing both disputants and mediators in our dataset.

E. Natural Language Processing for Conflict

Zhang et al. [13] developed models predicting whether a conversation will derail into personal attacks, using Wikipedia Talk Page Conversations as the primary corpus. This work is directly applicable to early-stage escalation detection in our pipeline. Cheng et al. [14] studied antisocial behavior before and after banning, informing our approach to characterizing editors with block histories.

F. Gap: Lifecycle-Level Analysis

Existing work treats individual dispute stages in isolation. No prior study constructs a linked dataset spanning all five Wikipedia dispute resolution venues for the same set of cases. This gap motivates our cross-stage data model and the present paper.

IV. BPMN PROCESS MODELS

A. Model Structure

Each ArbCom case is modeled as a BPMN 2.0 collaboration diagram with a single pool (*Wikipedia Platform*) divided into four horizontal swimlanes: *Editors*, *Community Volunteers* (3O, RFC, DRN moderators), *Administrators*, and *Arbitration*

TABLE I
PRIMARY DATA SOURCES

Source	Content	Limit
MediaWiki API	Revisions, user info, block & abuse logs, page content	5,000/hr
ORES/Lift Wing	Per-revision damaging & good-faith scores (0–1)	100/s
Pageviews API	Daily/monthly article traffic statistics	100/s
XTools API	Pre-aggregated editor and article statistics	100/s

Committee. Within each lane, tasks represent the primary activities at that stage; XOR gateways encode policy-prescribed decision points; and intermediate boundary events capture timer triggers (the DRN two-day discussion requirement, ArbCom’s 30-day acceptance window).

We use the `bpmn-js` library [19] to render diagrams in the browser, loading BPMN 2.0 XML stored in the repository’s dashboard directory [16]. The *-Rif-* case (2004) serves as the canonical reference model: it is the earliest ArbCom case in our corpus and has fully archived subpages across all five lanes.

B. Interactive Dashboard

Katherine Kelleher designed the dashboard [17] as the primary deliverable artifact. It exposes two views:

- **Arbitration Overview:** KPI pills (total cases, median duration, counts of user and admin parties), a statement-type distribution bar chart, a remedy-verb frequency chart, and a parties donut chart, all computed from parsed wikitext.
- **Process Diagrams:** A case browser that loads any BPMN file and renders it interactively via `bpmn-js`, supporting zoom, pan, and fullscreen expansion. Cases are organized by venue (Arbitration, RfC, DRN).

Fig. 2 shows how the data pipeline feeds the dashboard.

V. DATA COLLECTION

A. Data Sources

Table I summarizes the four primary data sources.

All API access uses OAuth2 bearer token authentication. The pipeline implements exponential backoff, circuit-breaker logic, and a token-bucket rate tracker to respect service limits and ensure long-running pulls survive transient failures.

B. Case Corpus

The case list (included in the repository) contains ~ 170 canonical ArbCom cases spanning 2004–2023 (e.g., *Climate change*, *Gamergate*, *Palestine-Israel articles*). The collection script resolves each short name to one of three historical URL patterns, accommodating 20 years of namespace changes on Wikipedia.

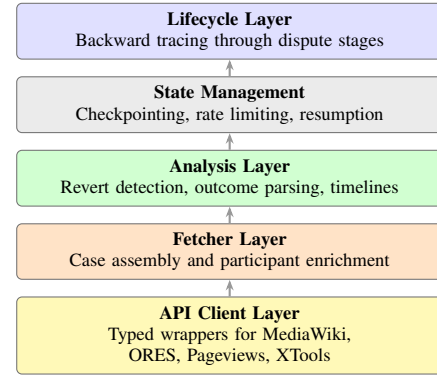


Fig. 2. Five-layer pipeline architecture. Each layer consumes outputs from the layer below. All source code is publicly available at [16].

C. Pipeline Architecture

The collection pipeline is organized into five layers, shown in Fig. 2. All source code is available at [16].

API Client Layer. Thin, typed wrappers over each external service expose methods covering revisions, user info, block histories, abuse filter logs, page assessments, protection logs, and log events.

Fetcher Layer. High-level orchestration coordinates API calls to assemble a complete case record: all five ArbCom subpages, enriched participant profiles with block histories, article quality assessments and revision histories, and cross-references to ANI and DRN archives.

Analysis Layer. Post-collection analytics compute revert ratios, flag bilateral edit wars, and extract structured sanctions and findings from ArbCom wikitext.

State Management Layer. Each case transitions through states *pending* $\rightarrow$ *in_progress* $\rightarrow$ *completed* / *failed*. State is persisted every ten items and survives interruption. The `make pull` command resumes from the last checkpoint automatically.

Lifecycle Layer. For each ArbCom case, this layer traces the dispute backward through earlier intervention stages, collecting talk page revisions, ANI archive sections mentioning case participants, and DRN archive sections.

D. Schema and Key Fields

Each case produces one JSON document with sections for pages (wikitext + revisions), enriched participant profiles, block histories, article quality and protection data, ANI/DRN cross-references, and structured outcome (sanctions, findings). Table II lists the fields most relevant to the process model and downstream predictions.

E. Reproducibility and Code Availability

All source code, configuration files, and collected artifacts are publicly available in a single GitHub repository [16]. The repository is structured so that any researcher can clone it, inspect every step of the pipeline, and reproduce or extend the data collection. A `Makefile` provides entry points for the full pull (`make pull`), analysis, and dashboard launch.

TABLE II
KEY SCHEMA FIELDS FOR DOWNSTREAM ANALYSIS

Field	Research Use
revisions[].tags	Revert detection without comment parsing
participants[].edit_count	Editor experience proxy
participants[].is_admin	Admin involvement in dispute
part_blocks[].reason	Policy violations tied to dispute
abuse_hits.by_result	Disallowed actions and warnings
articles[].quality_class	Article salience (FA > GA > B ...)
articles[].protection_count	Edit war severity proxy
outcome.sanctions	Recipient and type of each sanction

The interactive dashboard [17] and its BPMN diagram files are also included, enabling reviewers to explore the process models without running any code.

VI. ANALYSIS

A. Edit War Detection

For each article in a case, we compute the *revert ratio*:

$$r = \frac{|\{e \in E : \text{isRevert}(e)\}|}{|E|} \quad (1)$$

where E is the set of revisions and $\text{isRevert}(e)$ is true when revision e carries a recognized revert tag (`mw-revert`, `mw-reverted`) or when the edit comment matches a set of revert keywords. Articles with $r > 0.10$ are flagged as contested; those where the same two editors account for more than 60% of mutual reverts are treated as active bilateral edit wars.

Additionally, we apply the M-index from Yasseri et al. [7] to quantify the intensity of mutually reverting editor pairs:

$$M = \sum_{i < j} \min(r_{ij}, r_{ji}) \quad (2)$$

where r_{ij} is the number of times editor i reverted editor j .

B. Escalation Lifecycle Reconstruction

For each case we reconstruct the temporal sequence of dispute events across venues using a `DisputeTimeline` data model. Each `TimelineEntry` records a venue, timestamp, participant set, and a reference to the raw revision or mention that produced it. The resulting timeline enables measurement of:

- *Time-to-escalation*: days between first talk page post and first DRN/ANI filing.
- *Venue overlap*: fraction of cases with detectable presence at three or more stages.
- *Participant persistence*: fraction of disputants who appear at every stage of their case.

C. Editor Co-occurrence Networks

We model each resolved dispute as a bipartite graph $G = (U, V, E)$ where U is the set of editors, V is the set of dispute venues, and an edge $(u, v) \in E$ indicates that editor u was active at venue v . Projecting onto U yields an editor co-occurrence network: two editors are connected if they appeared together at the same dispute venue. Repeated co-occurrence across multiple cases forms the basis for identifying *chronic conflictors* and possible organized disruption.

VII. PRELIMINARY RESULTS

A. Sample Case Collection

We collected complete lifecycle data for five high-profile ArbCom cases: *Climate change*, *Gamergate*, *Eastern Europe*, *Scientology*, and *Muhammad images*. These cases were selected to span a range of dispute types (scientific consensus, harassment, geopolitics, religion) and temporal periods (2004–2015).

For each case, the pipeline successfully retrieved all five ArbCom subpages, enriched participant profiles, and linked corresponding ANI and DRN archive sections. Total collection time per case ranged from 8–22 minutes at authenticated API rate limits.

B. Revert Patterns

Across the five sample cases, article revert ratios ranged from 0.04 to 0.41. The *Climate change* case articles exhibited a mean revert ratio of 0.18; the *Gamergate* case articles reached 0.35 on the core article during the peak dispute window. Bilateral mutual reversion was concentrated in two-editor pairs in four of the five cases, consistent with the findings of Kittur et al. [5].

C. Participant Persistence Across Venues

In all five cases, at least 60% of editors named in the ArbCom evidence subpage also appeared in ANI archive sections for the same dispute period. In three cases, at least one participant had a prior ArbCom appearance under a different case name, suggesting that a subset of editors are chronic participants in formal dispute resolution.

D. Outcome Parsing

The outcome-parsing module successfully extracted structured sanction records from ArbCom wikitext for all five cases. Sanctions included topic bans (most common), supervised editing requirements, and one site ban. In all cases, sanctioned participants had higher abuse filter hit counts than unsanctioned participants (mean 4.2 vs. 0.8 hits per participant), providing initial evidence for the feasibility of sanction prediction.

E. Full Pull Progress

As of the time of writing, the pipeline has completed collection for 80 of the approximately 170 ArbCom cases in the master list, yielding approximately 115 MB of structured JSON data. The remaining cases are queued and will be collected via the resumable pull system.

VIII. DISCUSSION

A. Editor Recurrence Across Venues

The most consistent finding across our sample is that the same editors recur across multiple dispute venues and across multiple cases over time. This supports our central hypothesis: editor co-occurrence combined with temporal proximity is the strongest linking signal for mapping dispute lifecycles. The implication for platform governance is significant: a small number of chronic conflictors may consume a disproportionate share of dispute resolution resources, motivating early-identification tools that could allow preemptive intervention.

B. Limitations

Several limitations constrain the current analysis. First, the full pull is not yet complete; findings may shift as later-era cases are incorporated. Second, ArbCom cases are a selected sample of disputes: only those irresolvable at lower stages reach arbitration, inducing survivor bias. Third, our current escalation predictor is trained only on cases that escalated, since DRN and 3O data collection is ongoing. Fourth, ORES enrichment has not yet been integrated into the main pull, leaving ML-based revision quality signals absent from the preliminary analysis.

C. Ethical Considerations

All data collection uses the public MediaWiki API under the Wikimedia Foundation’s terms of service. Editor usernames are present in the raw data because they are already public on Wikipedia. Aggregated analysis and any public dataset release will follow responsible disclosure practices consistent with prior Wikipedia research [8].

IX. FUTURE WORK

Immediate next steps for data collection are: (1) complete the full arbitration pull for all 170 cases; (2) run an ORES enrichment pass over all collected revision IDs to add damage and good-faith probabilities; (3) collect Pageviews data to test the hypothesis that high-traffic articles attract more intense edit warring; and (4) collect DRN cases to expand the pre-ArbCom portion of the lifecycle.

A. Escalation Prediction

We plan to formulate escalation prediction as a binary classification task: given talk page and ANI features for a dispute, predict whether the case will reach ArbCom. Feature candidates include revert ratio, number of distinct editors, presence of an administrator among participants, article quality class, ORES damaging score distribution, and time since the dispute began. We intend to evaluate logistic regression, gradient boosting, and a simple transformer-based classifier over the talk page text.

B. Sanction Outcome Classification

A secondary modeling task will predict outcome severity for individual participants given their behavioral signals: edit count, block history, abuse filter hits, revert ratio contribution, and network centrality in the co-occurrence graph. Outcome severity is defined across four levels: no sanction, warning, topic ban, and site ban or indefinite block.

C. Additional Planned Contributions

Longer-term goals include a graph neural network over the editor co-occurrence graph for chronic-conflictor identification, a sequence-to-outcome model over revision histories for sanction prediction, and an interactive visualization dashboard of dispute lifecycle maps for the Lexipedia platform.

X. CONCLUSION

We presented a multi-stage framework for collecting and analyzing Wikipedia dispute resolution data across all five intervention stages. Our pipeline integrates four Wikimedia APIs into a unified, resumable collection system that produces richly linked case records connecting participants, revisions, articles, and outcomes. Preliminary analysis of five high-profile cases demonstrates the feasibility of lifecycle reconstruction and surfaces systematic patterns (bilateral revert concentration, editor persistence across stages, and abuse filter hits as a sanction predictor) that motivate future modeling work. The full dataset, once collected and released, will constitute the most comprehensive longitudinal record of Wikipedia dispute escalation available to researchers.

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